

Savitribai Phule Pune University, Pune Third Year Information Technology (2019 Course) 314454 (C): Elective-II-(Cloud Computing)		
Teaching Scheme:	Credit Scheme:	Examination Scheme:
Theory (TH) : 3 hrs/week	03 Credit	Mid_Semester : 30 Marks End_Semester : 70 Marks
Prerequisite Courses: 1. Basics of Computer Networks 2. Operating Systems		
Course Objectives: 1. To provide students with the fundamentals and essentials of cloud computing 2. To learn basics of virtualization and its importance 3. To provide students a sound foundation of the cloud computing so that they are able to start using and adopting cloud computing services and tools in their real life scenarios 4. To enable students exploring some important cloud computing driven commercial systems and applications 5. To understand cloud storage technologies and relevant file systems 6. To be exposed to Ubiquitous Cloud and Internet of Things		
Course Outcomes: On completion of the course, students will be able to– CO1: Articulate the main concepts, key technologies and fundamentals of cloud computing. CO2: Understand cloud enabling technologies and virtualization. CO3: Analyze various cloud programming models and apply them to solve problems on the cloud. CO4: Explain data storage and major security issues in the cloud. CO5: Understand trends in ubiquitous cloud and internet of things. CO6: Explore future trends of cloud computing.		
COURSE CONTENTS		
Unit I	FUNDAMENTALS OF CLOUD COMPUTING	(06 hrs)
Origins and Influences, Basic Concepts and Terminology, Goals and Benefits, Risks and Challenges, Roles and Boundaries, Cloud Characteristics, Cloud Delivery Models, Cloud Deployment Models, Federated Cloud/Intercloud, Types of Clouds.		
Mapping of Course Outcomes for Unit I	CO1	
Unit II	CLOUD-ENABLING TECHNOLOGY AND VIRTUALIZATION	(06 hrs)

Cloud-Enabling Technology: Broadband Networks and Internet Architecture, Data Center Technology, Virtualization Technology, Web Technology, Multitenant Technology, Service Technology.

Implementation Levels of Virtualization, Virtualization Structures/Tools and Mechanisms, Types of Hypervisors, Virtualization of CPU, Memory, and I/O Devices, Virtual Clusters and Resource Management, Virtualization for Data-Center Automation.

Mapping of Course Outcomes for Unit II

CO2

Unit III

COMMON STANDARDS AND CLOUD PLATFORMS

(06 hrs)

Common Standards: The Open Cloud Consortium, Open Virtualization Format, Standards for Application Developers: Browsers (Ajax), Data (XML, JSON), Solution Stacks (LAMP and LAPP), Syndication (Atom, Atom Publishing Protocol, and RSS), Standards for Security.

Amazon web services: Compute services Storage Services Communication Services Additional services

Google AppEngine: Architecture and core concepts, Application life cycle, Cost model

Microsoft Azure: Azure core concepts, SQL Azure, Windows Azure platform appliance

Mapping of Course Outcomes for Unit III

CO3

Unit IV

DATA STORAGE AND SECURITY IN CLOUD

(06 hrs)

Cloud file systems: GFS and HDFS, BigTable, HBase and Dynamo Cloud data stores: Datastore and Simple DB Gautam Shrauf, Cloud Storage-Overview, Cloud Storage Providers.

Securing the Cloud- General Security Advantages of Cloud-Based Solutions, Introducing Business Continuity and Disaster Recovery. Disaster Recovery- Understanding the Threats.

Mapping of Course Outcomes for Unit IV

CO4

Unit V

UBIQUITOUS CLOUDS AND THE INTERNET OF THINGS

(06 hrs)

Cloud Trends in Supporting Ubiquitous Computing, Performance of Distributed Systems and the Cloud, Enabling Technologies for the Internet of Things (RFID, Sensor Networks and ZigBee Technology, GPS), Innovative Applications of the Internet of Things (Smart Buildings and Smart Power Grid, Retailing and Supply-Chain Management, Cyber-Physical System), Online Social and Professional Networking.

Mapping of Course Outcomes for Unit V

CO5

Unit VI

FUTURE OF CLOUD COMPUTING

(06 hrs)

How the Cloud Will Change Operating Systems, Location-Aware Applications, Intelligent Fabrics, Paints, and More, The Future of Cloud TV, Future of Cloud-Based Smart Devices, Faster Time to Market for Software Applications, Home-Based Cloud Computing, Mobile Cloud, Autonomic Cloud Engine, Multimedia Cloud, Energy Aware Cloud Computing, Jungle Computing. **Docker at a Glance:** Process Simplification, Broad Support and Adoption, Architecture, Getting the Most from Docker, The Docker Workflow

Mapping of Course Outcomes for Unit VI	CO6
Text Books:	
1. Thomas Erl, Zaigham Mahmood and Ricardo Puttini, Cloud Computing: Concepts, Technology & Architecture, Pearson, ISBN :978 9332535923, 9332535922, 1 st Edition 2. Anthony T. Velte Toby J. Velte, Robert Elsenpeter, "Cloud Computing: A Practical Approach", 2010, The McGraw-Hill.	
Reference Books:	
1. Rajkumar Buyya, Christian Vecchiola, S. ThamaraiSelvi, Mastering Cloud Computing: Foundations and Applications Programming, McGraw Hill, ISBN: 978 1259029950, 1259029956. 2. Gautam Shrof, "ENTERPRISE CLOUD COMPUTING Technology Architecture, Applications, Cambridge University Press, ISBN: 9780511778476 3. Srinivasan, J. Suresh, Cloud Computing: A practical approach for learning and implementation, Pearson, ISBN :9788131776513. 4. Jack J. Dongarra, Kai Hwang, Geoffrey C. Fox, Distributed and Cloud Computing: From Parallel Processing to the Internet of Things, Elsevier, ISBN :9789381269237, 9381269238, 1st Edition. 5. Brian J.S. Chee and Curtis Franklin, Jr., Cloud Computing: Technologies and Strategies of the Ubiquitous Data Center, CRC Press, ISBN :9781439806128. 6. Kris Jamsa, Cloud Computing: Saas, Paas, Iaas, Virtualization, Business Models, Mobile, Security, and More, Jones and Bartlett, ISBN :9789380853772. 7. John W. Rittinghouse, James F. Ransome, Cloud Computing Implementation, Management, and Security, CRC Press, ISBN : 978 1439806807, 1439806802. 8. Karl Matthias, Sean P. Kane, Docker: Up and Running, O'Reilly, ISBN:9781491917572, 1491917571. 9. Barrie Sosinsky, Cloud Computing Bible, Wiley, ISBN: 978 8126529803. 10. Ronald L. Krutz and Russell D. Vines, Cloud Security: A Comprehensive guide to Secure Cloud Computing, Wiley, ISBN: 9788126528097. 11. Scott Adkins, John Belamaric, Vincent Giersch, Denys Makogon, Jason E. Robinson, OpenStack: Cloud Application Development, Wrox, ISBN :9781119194316. 12. Kailash Jayaswal, Jagannath Kallakurchi, Donald J. Houde, Cloud Computing Black Book ,Wiley Dreamtech, ISBN:9789351194187	